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GAN-based SAR and optical image translation for wildfire impact assessment using multi-source remote sensing data

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ABSTRACT

Despite the popularity and success in burned area detection and assessment, multispectral satellite images are often affected by poor sunlight-illumination conditions, particularly at high latitudes. Given that Synthetic Aperture Radar (SAR) can effectively penetrate clouds and collect images in all-weather conditions during day and night, the complementary use of optical and SAR data can be helpful for remote-sensing measurements and assessments of burned sites. Nevertheless, the widely used burn-sensitive spectral indices hardly help analyze SAR data due to the inherent difference between optical and SAR sensors in physical imaging mechanisms. In this study, we aim to leverage multi-source data for burned area mapping and burn severity assessment by translating SAR images into optical images using ResNet-based Pix2Pix model. Experiments were performed on 8669 pairs of bitemporal Sentinel-1 SAR and Sentinel-2 optical patches over 304 large wildfire events in Canada with a wide range of land covers. The translated optical images from SAR data occupied similar spectral properties to real optical observations and the corresponding generated spectral indices (*i.e.*, delta Normalized Burn Ratio (dNBR), relative dNBR, and Relativized Burn Ratio) also showed high agreement with real optical indices. In terms of burned area detection using the generated indices, their medium values of the area under the receiver operating characteristics curve (AUC) were over 85%, achieving promising performance that outperformed the SAR-based index. Burn severity maps derived from multi-source data achieved a relatively high Kappa coefficient of 0.77. Results showed the feasibility and effectiveness of GAN-based SAR-to-optical translation for wildfire impact assessment, paving the way for the combined use of optical and SAR data.

1. Introduction

Wildfires attract more research interest due to their increasing frequency around the globe even at high northern latitudes (*e.g.*, Alaska and Canada) (Bowman et al., 2017). Boreal forests and peatlands cool the climate but the fire-induced disturbances would lead to significant loss of vegetation and an increase in greenhouse gas emission from North American wildland fires (French et al., 2011). Earth Observation satellites offer a unique opportunity for mapping and assessing burned areas at regional and global scales (Chuvieco and Congalton, 1988; Chuvieco et al., 2019, 2020). Moreover, recent advancements in artificial intelligence, new generations of satellites, and extensive applications of remote sensing techniques provide promising tools for systematic monitoring and inventory of wildfires on Earth with enhanced revisit frequency and improved spatial resolution (Chuvieco et al., 2019; Yuan et al., 2020).

Global burned area products (*e.g.*, MCD64A1 c6, MCD14ML, and FireCCI51) were derived primarily based on NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) sensors, providing high temporal resolution and coarse spatial resolution from 250 m to 1 km (Justice et al., 2002; Giglio et al., 2018; Humber et al., 2019; Chuvieco et al., 2019; Lizundia-Loiola et al., 2020). To detect small burned areas (<100 ha), medium-resolution multispectral satellite data stands out as one of the primary sources for burned area detection with higher spatial resolution than coarse MODIS-like data. Landsat series data and Sentinel-2 data have been extensively exploited for mapping burned areas and burn severity levels. Bastarrika et al. (2011) classified burned pixels from Landsat Thematic Mapper (TM)-Enhanced Thematic Mapper plus (ETM+) images using spectral indices and a region-growing algorithm to balance omission and commission errors. Sentinel-2 data improved the accuracy of the small burn scar database at 20-m spatial resolution across sub-Saharan Africa in comparison to MODIS burned area product (500 m for MCD64A1 c6) (Roteta et al., 2019). Based on multispectral satellite data, burn-sensitive spectral indices were proposed such as delta normalized burn ratio (dNBR) (Lutes et al., 2006), relative dNBR (RdNBR) (Miller and Thode, 2007), and relativized burn ratio (RBR_{opt}) that typically have a high correspondence

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to field-based burn severity measurements with R^2 over 0.65 (Parks et al., 2014). Landsat TM/ETM+ imagery was utilized to quantify burn severity based on dNBR and RdNBR over boreal forests in western Canada (Soverel et al., 2010). Additionally, Quintano et al. (2018) compared the classification accuracy of burn severity levels over a large wildfire in Spain based on dNBR, RdNBR, and RBR_{opt} using Landsat-8 and Sentinel-2 data. Mallinis et al. (2018) demonstrated that dNBR showed slightly higher classification accuracy for Sentinel-2A than Landsat-8 in Mediterranean pine forest areas. On the other hand, optical remote sensing data can be limited by the temporal coverage (*e.g.*, 5 days for Sentinel-2A/B and 16 days for Landsat-8), bad weather or potential cloud contamination (Belenguer-Plomer et al., 2019). Synthetic Aperture Radar (SAR) data can compensate for the lack of cloud-free optical images over fire-prone regions.

SAR has become a suitable choice for burned area detection by taking advantage of its certain penetration capacity, improved spatial resolution, increased data availability, and sensitivity to vegetation structural changes (Tanase et al., 2015; Belenguer-Plomer et al., 2019). Burned surfaces have various radar signatures that depend on the radar wavelength polarization, burn severity level on trees, and environmental impacts (*e.g.,* soil moisture) (Belenguer-Plomer et al., 2019). The main idea of SAR-based burned area detection is to interpret radar backscatter signal behavior to burn scars (Zhang et al., 2019; Ban et al., 2020). The possibility of estimating wildfire impacts using radar data (*e.g.,* SAR backscatter intensity, interferometric coherence, and polarimetric decomposition components) has been demonstrated by Gimeno et al. (2004), Huang and Siegert (2006), and Tanase et al. (2010, 2014, 2015). Radar Burn Ratio (RBR_{sar}) is a typical index for burn severity estimation by computing the post- to pre-fire ratio of the backscatter or radar based index (Tanase et al., 2015; Fernandez-Carrillo et al., 2019). RBR_{sar} exhibits a good agreement with the in-situ reference data in burn severity assessment. Even though Sentinel-1 SAR data has acceptable performance, optical-based indices provide more accurate burned area detection (Dice coefficient 0.68 *vs*. 0.45) (Tanase et al., 2020). Some studies combined the optical and SAR data for burned area detection using deep learning-based algorithms. Belenguer-Plomer et al. (2021) demonstrated that SAR and optical data integration only slightly improved or maintained the accuracy in comparison to the optical-based results. Zhang et al. (2021) found that the synergy of SAR backscatter coefficient and optical surface reflectance achieved lower accuracy than optical data alone for burned area mapping. The disparate imaging mechanisms of SAR and optical data limit the performance of these purely data-driven methods to fuse multi-source images. On the other hand, the increased computational complexity of SAR data interpretation and the large number of samples produced by interferometric techniques or time-series SAR data limit the generalization of deep learning approaches (Zhang et al., 2021). A new promising idea is to fuse SAR and optical images by utilizing multi-source data within the same domain.

As a breakthrough in deep learning, Generative Adversarial Networks (GAN) learn the internal data distribution from abundant data and generate new synthetic instances of data (Goodfellow et al., 2014). Conditional GAN (cGAN) was proposed to control the data generation process and was applied to many image-to-image architectures such as Pix2Pix (Zhu et al., 2017) and Cycle-Consistent GAN (Zhu et al., 2017). To fill the gap between SAR and optical images, GAN has been used to translate SAR to optical images (Choi et al., 2018; Reyes et al., 2019; Li et al., 2020b; Gao et al., 2020; Li et al., 2021), produce samples for SAR-to-optical matching (Hughes et al., 2018; Merkle et al., 2018) and SAR image classification (Li et al., 2020a; Ren et al., 2020). Bermudez et al. (2019) applied a GAN-based SAR image synthesis method to classify burned areas but it additionally requires multi-temporal data as conditioning variables, which might be inaccessible in real scenarios.

This work investigates the possibility of the combined use of paired Sentinel-1 SAR and Sentinel-2 optical images for assessing wildfire impacts over hundreds of major wildfire events in Canada from 2017 and 2018. The objectives are three-fold. First, we aim to propose a novel framework that can exploit both SAR and optical data for burned area mapping and burn severity assessment when optical observations are not available (*e.g.*, for areas with persistent cloud cover or thick smoke), by translating SAR images into optical images. Second, we desire to further stabilize the training of the SAR-to-optical translation model and improve the diversity of the generated images using an improved Pix2Pix model with structural similarity index (SSIM) reconstruction loss and LSGAN (Least Squares GAN) adversarial loss. Third, we would like to investigate the difference in performance of multi-sensor indices in detecting burned areas and mapping burn severity in comparison to optical-based or SAR-based indices. In sum, the main objective of this work is to investigate the possibility of translating SAR to optical images for the usability of classical spectral indices in wildfire impact assessment.

2. Study area and data

2.1. Study area

The study area covers eight fire-prone provinces/territories in Canada as shown in Fig. 1(a). The burn perimeters were acquired from the National Burned Area Composite (NBAC) data archive published by Nature Resources Canada (2021). They were rasterized as reference masks to define the Region of Interest (ROI) and compute burned areas in km^2. Fig. 1(b) shows the land cover distribution of our study area. The dominant land cover type is the closed forest with evergreen needle leaf, while the open forest (class values: 121, 126) occupies about 18.5%. For the detailed description of the land cover, please refer to Copernicus Global Land Cover Layers.[1]

2.2. Data construction

Sentinel-2 Multispectral Instrument (MSI) Level 1-C images and Sentinel-1 SAR Ground Range Detected dual-polarized (vertical–vertical VV, and vertical–horizontal VH polarization) images in interferometric wide swath mode were used. We produced a SAR-optical multi-source data set including bi-temporal Sentinel-1 SAR and Sentinel-2 MSI images based on NBAC burned area perimeters and Google Earth Engine (Gorelick et al., 2017). For each wildfire event, an ROI was defined according to its perimeter bounding box buffered by 2000 m. The same fire period one year before and one year after the fire year was taken as the pre-fire period and post-fire period respectively. For Sentinel-1 SAR data, the average image of all SAR images acquired on the same orbits and in the pre-fire period was taken as a pre-fire SAR image. Similarly, the average image of all SAR images was acquired in the post-fire period as a post-fire SAR image. It can minimize the seasonal difference between pre- and post-fire images. On the other hand, the Sentinel-2 images acquired in the pre-fire period and the post-fire period were mosaiced as a pre-fire optical image and a post-fire optical image after being sorted by the "CLOUDY_PIXEL_PERCENTAGE" property. For the pre- and post-fire SAR images with VV and VH bands, Normalized Difference Backscatter Intensity (NDBI) was computed in dB as the third band of SAR images ($\mathrm{NDBI} = 10\log_{10}(\frac{VV-VH}{VV+VH})$).

After data preprocessing, the coregistered Sentinel-1 pre- and post-fire SAR images, Sentinel-2 pre- and post-fire MSI images, and the reference mask were exported over the ROI of each fire event at 20 m spatial resolution. All exported images were visually inspected and cleaned by removing some scene pairs if either of them contained large no-data areas, large clouds, or mosaicing artifacts. In total, there are 304 wildfire events. 281 events are used for training and 23 for testing. These coregistered scenes were tiled into patches of shape

[1] https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_Landcover_100m_Proba-V-C3_Global#bands

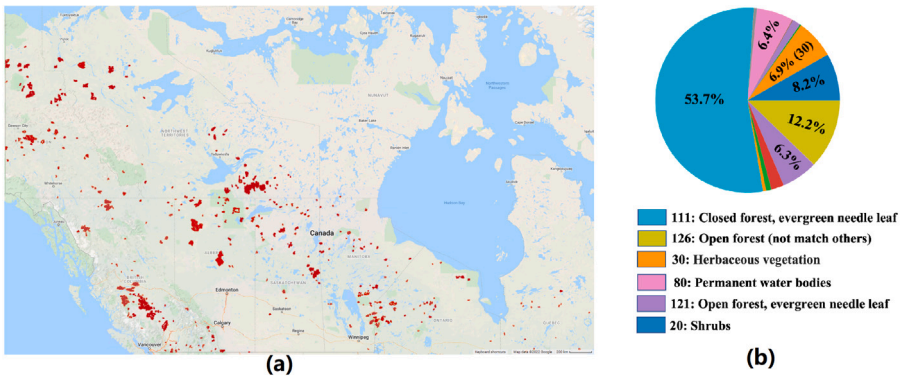

Fig. 1. (a) Spatial coverage of the study area. (b) Land cover distribution over the study area.

Table 1
Study area and data. Provincial acronyms: AB = Alberta, BC = British Columbia, MB = Manitoba, ON = Ontario, QC = Quebec, and SK = Saskatchewan. Territorial acronyms: NT = Northwest Territory and YT = Yukon Territory.

	Training data			Testing data			
	No. patches	Burned area (km^2)	Unburned area (km^2)	Fire event ID (Fire period: start date-end date)	No. patches	Burned area (km^2)	Unburned area (km^2)
AB	291	749	6880	CA_AB_2017_2418 (2017/8/31-2017/9/22) CA_AB_2017_532 (2017/8/13-2017/9/15)	55	345	983
BC	3170	18,833	64,267	CA_BC_2018_182 (2018/7/27-2018/9/15) CA_BC_2018_267 (2018/8/1-2018/9/8) CA_BC_2017_2418 (2017/8/29-2017/9/17)	305	2246	5231
MB	927	2806	21,495	CA_MB_2018_1090 (2018/6/28-2018/7/27) CA_MB_2018_330 (2018/7/12-2018/8/30) CA_MB_2018_529 (2018/5/18-2018/6/4) CA_MB_2017_1332 (2017/8/16-2017/9/23)	218	1298	3349
NT	724	2095	16,884	CA_NT_2017_126 (2017/7/14-2017/8/12) CA_NT_2017_17 (2017/7/28-2017/8/20) CA_NT_2017_48 (2017/7/5-2017/9/3)	114	737	1895
ON	908	3363	20,440	CA_ON_2018_1667 (2018/7/9-2018/8/5) CA_ON_2018_529 (2018/5/21-2018/8/30)	72	478	1049
QC	275	612	6597	CA_QC_2018_1258 (2018/6/28-2018/7/8) CA_QC_2018_322 (2018/6/29-2018/7/7) CA_QC_2018_838 (2018/6/28-2018/7/8)	44	180	662
SK	683	2945	14,960	CA_SK_2017_1033 (2017/8/4-2017/9/26) CA_SK_2017_592 (2017/8/10-2017/9/19) CA_SK_2017_617 (2017/8/3-2017/9/19)	188	1180	3061
YT	580	2646	12,558	CA_YT_2017_43 (2017/6/18-2017/8/2) CA_YT_2017_60 (2017/7/8-2017/8/26)	115	533	2225
	7558	34,048	164,080		1111	6998	18,453

$c \times 256 \times 256$ (c denotes the number of channels) in a stride of 256 pixels, without overlapping pixels. After producing the raw MSI and SAR patches, three bands (B12, B11, and B8 A) of each Sentinel-2 image patch were selected to construct a false-color composition. Similarly, VV and VH backscatter values (*i.e.*, σ^0_{VV} and σ^0_{VH}) and NDBI were used to create a false-color composition (NDBI-VH-VV in RGB). Eventually, our dataset consists of 8669 pairs of pre- and post-fire SAR-optical patches, in which 7558 pairs are for training and 1111 pairs for testing, as shown in Table 1. The fire ID of each test scene in Table 1 also exhibits the administrative regions where training and testing patches were sampled from.

3. Methodology

The different imaging mechanisms between optical and SAR sensors are bound to introduce unavoidable errors in detecting burned pixels if optical-based algorithms are applied directly to SAR data. This study aims to translate SAR images to optical images to adjust for the usability of classical methods in wildfire impact assessment. Towards this goal, we proposed a framework that consists of three parts, an image-to-image translation network, *e.g.,* Pix2Pix (Zhu et al., 2017), a spectral indices module, and a wildfire impact assessment process. As shown in Fig. 2(a), one of the heterogeneous images (*i.e.,* Sentinel-1 SAR) is first translated to the optical domain (*i.e.,* Sentinel-2 MSI) by the trained Pix2Pix model on the coregistered SAR-optical multi-source dataset. Then the generated optical image and the other available optical image (or two generated optical images) are utilized to derive bi-temporal indices, such as dNBR, RdNBR, and RBR_{opt} in Fig. 2(b). Subsequently, we investigate the practical applications such as burned area detection and burn severity assessment using the generated indices in Fig. 2(c).

3.1. ResNet-based Pix2Pix translation network

GAN-based image-to-image translation has received more attention since Pix2Pix appeared (Zhu et al., 2017), which learns from the sufficient paired training data. The image translation network in this study aims at diminishing discrepancies between optical and SAR images by transforming them into the same domain and providing explainable input for spectral indices computation. During the adversarial training phase, SAR and optical image pairs are fed into the ResNet-based Pix2Pix network. The residual generator takes SAR images as input and outputs optical images. The PatchGAN discriminator attempts to judge

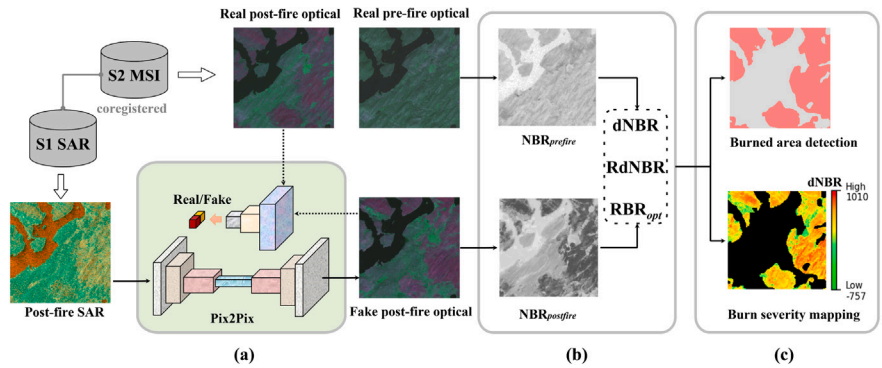

Fig. 2. The structure of the proposed framework with a three-step process. (a) Pix2Pix translation network. (b) Spectral indices computation module. (c) Applications on burned area detection and burn severity mapping.

whether translated outputs look like the corresponding real optical data. The regular Pix2Pix model by Zhu et al. (2017) took a typical UNet as the generator network with vallina and L1 loss terms. In this study, we replaced the UNet with a residual generator to achieve better performance based on LSGAN and SSIM norms.

3.1.1. Residual generator and patchgan discriminator

The deep residual networks (ResNets) noticeably increase the depth of a network (Xie et al., 2017), leading to improved performance in numerous applications such as image recognition, image classification, image generation, image reconstruction, and image super-resolution (He et al., 2016; Li et al., 2019; Park and Shin, 2022; Du et al., 2020; Muhammad Umer et al., 2020). Fig. 3(a) demonstrates the workflow of the residual generator. The input and output are SAR images and generated optical images with the size of 3 × 256 × 256, respectively. The encoder takes the input image and produces a high-level feature vector during downsampling. The feature vector is then passed to six stacking residual blocks. Two transpose convolution layers in the upsampling process are applied to decoder features from residual blocks and produce the reconstructed image. In Fig. 3(a), we omit the instance normalization and activation layer after each layer in the downsampling and upsampling process for simplicity. The residual generator can handle images of arbitrary size once trained owing to its fully convolutional architecture.

The discriminator used in this study is a PatchGAN with four layers (Zhu et al., 2017). Enabling faster training with fewer parameters, PatchGAN is more computationally efficient against the global inference over the entire input. Additionally, PatchGAN focuses more on surface features like texture and style in the translation task. In this study, the input images of the discriminator are a real SAR image and a translated/real optical image as shown in Fig. 3(b). The convolutional layers of PatchGAN downsample the feature maps and double the number of feature channels. The output of the discriminator is a 30 × 30 probability map.

3.1.2. Loss function

ResNet-based Pix2Pix adopts adversarial loss and reconstruction loss in SAR-to-optical image translation. Adversarial loss penalizes the generator for yielding more realistic optical images, namely, it forces the generated images to be indistinguishable from the real optical images to fool the discriminator. The reconstruction loss aims to produce a realistic image near the target domain (*i.e.*, a real optical image). The adversarial loss in the regular Pix2Pix, namely vanilla loss can be represented as the following min–max optimization formulation:

$$\min_G \max_D \mathcal{L}_{vanilla}(G, D) = \mathbb{E}_{x,y}[\log D(x, y)] + \mathbb{E}_{x,z}[\log(1 - D(x, G(x, z)))] \tag{1}$$

where x and y represent the input paired SAR image and optical image, respectively. z noise is represented in the form of dropout. $G(x, z)$ denotes the generated optical image.

Vanilla loss in regular GANs may suffer from vanishing gradient problems caused by the sigmoid cross entropy loss function (Mao et al., 2019). To address this issue, Mao et al. (2017) proposed the least-squares loss function for the GAN discriminator (*i.e.*, LSGAN), which shows more stable training and produces better images. LSGAN forces the generator to generate samples towards the decision boundary. Along with that, it can overcome the vanishing gradient issue as the samples that are far away from the real data generate more gradients while updating the generator. The following is the objective function for LSGAN:

$$\begin{aligned} \min_D V_{\text{LSGAN}}(D) &= \frac{1}{2}\mathbb{E}_{x,y}\left[(D(x, y) - b)^2\right] + \frac{1}{2}\mathbb{E}_{x,z}\left[(D(x, G(x, z)) - a)^2\right] \\ \min_G V_{\text{LSGAN}}(G) &= \frac{1}{2}\mathbb{E}_{x,z}\left[(D(x, G(x, z)) - c)^2\right] \end{aligned} \tag{2}$$

where a is the label for the fake sample and it is set as -1; b is the label for the real sample as 1; c represents the degree that the generator wants the discriminator to believe the fake sample. c can be set as 0. Hence, we have two individual loss functions that are being optimized. The model distribution can be learned through this adversarial min–max optimization, where the generator aims to minimize it against the discriminator that wants to maximize it (Miyato et al., 2018).

Regarding the reconstruction loss, the L1 loss used in the regular Pix2Pix can be expressed as:

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z}\left[\|y - G(x, z)\|_1\right] \tag{3}$$

where $\|.\|_1$ represents L1 distance. ResNet-based Pix2Pix in this study replaces L1 loss with SSIM loss (Zhao et al., 2016). SSIM loss is perceptually motivated and penalizes the generator to produce visually pleasing images with good structural similarity. $\mathcal{L}_{\text{SSIM}}$ denotes the SSIM loss between y and $G(x, z)$, defined as:

$$\mathcal{L}_{\text{SSIM}}(G) = \frac{1}{N}\sum_{p\in y, p\in G(x,z)} 1 - \frac{2\mu_m\mu_n + C_1}{\mu_m^2 + \mu_n^2 + C_1} \cdot \frac{2\sigma_{mn} + C_2}{\sigma_m^2 + \sigma_n^2 + C_2} \tag{4}$$

where $m(p)$ and $n(p)$ are pixels located at p in y and $G(x, z)$, respectively; μ_m and μ_n represent the mean values of pixels in y and $G(x, z)$. σ_n and σ_m denote the standard deviation values of pixels in y and $G(x, z)$. σ_{mn} is the covariance of pixels in y and $G(x, z)$. N is the total number of pixels in y or $G(x, z)$. C_1 and C_2 are constants to stabilize the division with a weak denominator.

In this study, the objective function to update the weights of the discriminator and generator can be defined as:

$$\arg\min_G \max_D \mathcal{L}_{LSGAN}(G, D) + \lambda\mathcal{L}_{SSIM}(G) \tag{5}$$

where the first term represents the LSGAN loss and the second part denotes the SSIM loss of the generator. λ is a regularization weight that controls the weight of the SSIM loss set as 1.0.

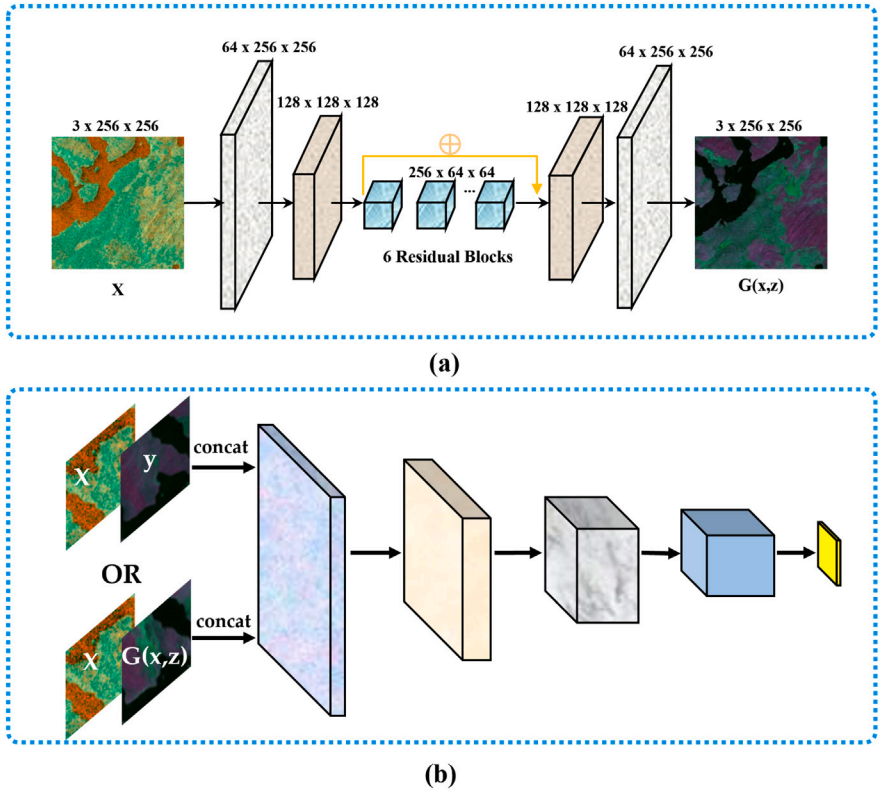

Fig. 3. (a) The architecture of ResNet-based Pix2Pix generator. (b) The architecture of PatchGAN discriminator.

3.1.3. Network training

The generator and discriminator are alternatively optimized by applying a dynamic iterative training strategy. When the parameters of the generator are fixed, the discriminator tries to distinguish between generated optical images and real images by labeling them as fake and real, respectively. On the other hand, the generator is optimized by making the generated optical images indistinguishable from the real images thus labeling them as real in the discriminator with the fixed parameters. The training of the generator and discriminator network carefully controls and balances each other by alternately optimizing the adversarial objective function (Eq. (5)).

3.2. Spectral and SAR-based metrics calculation

Using the trained ResNet-based Pix2Pix model, the generator can translate a SAR image to an optical image, which is then used to calculate spectral indices with the other available optical image. It should be noted that the pre- and post-fire images used for the bi-temporal indices computation can both be generated from their corresponding SAR images if no cloud-free optical images are available.

The most commonly used spectral indices in fire science are dNBR and its relativized forms (RdNBR, RBR_{opt}), all of which rely on the normalized burn ratio (NBR) between pre- and post-fire scenes as follows:

$$NBR = \frac{NIR - SWIR2}{NIR + SWIR2} \tag{6}$$

$$dNBR = \left(NBR_{\text{prefire}} - NBR_{\text{postfire}}\right) \times 1000 \tag{7}$$

$$RdNBR = \begin{cases} \frac{dNBR}{\left|NBR_{\text{prefire}}\right|^{0.5}}, \left|NBR_{\text{prefire}}\right| \geq 0.001 \\ \frac{dNBR}{|0.001|^{0.5}}, \left|NBR_{\text{prefire}}\right| < 0.001 \end{cases} \tag{8}$$

$$RBR_{opt} = \frac{dNBR}{NBR_{\text{prefire}} + 1.001} \tag{9}$$

where near-infrared (NIR) and shortwave-infrared 2 (SWIR2) in Eq. (6) are the Sentinel-2 MSI band B8A and band B12, respectively. dNBR (Eq. (7)) measures the absolute change between pre- and post-fire NBR values, which limits the ability to quantify the high burn severity. For example, low pre-fire vegetation cover will generally produce low dNBR values even though pre-fire vegetation is largely consumed. The denominator in the Eq. (8) allows RdNBR to be more sensitive than dNBR to vegetation changes in that case. Theoretically, the relative index like RdNBR could detect burned areas or assess burn severity at a consistent scale but it may fail if pre-fire NBR values are extremely low. As an alternative relativized metric, RBR_{opt} (Eq. (9)) is proposed to avoid infinity and failing under any circumstances by adding 1.001 to the denominator (Parks et al., 2014).

In terms of the SAR-based metric, RBR_{sar} can be computed to evaluate the burn severity without prior information, which exhibits the pre- and post-fire SAR backscatter variants attributed to structural plant damages (Tanase et al., 2015):

$$RBR_{xy} = \frac{\gamma^0\text{postfire}_{xy}}{\gamma^0\text{prefire}_{xy}} \tag{10}$$

where xy refers to the polarization mode (*i.e.,* VV or VH for Sentinel-1 SAR data) or NDBI and γ^0 (gamma nought) is the normalized backscatter coefficient in linear scale (Fernandez-Carrillo et al., 2019).

3.3. Evaluation metrics

We assess the performance of each module both qualitatively and quantitatively on evaluation metrics. Four image quality assessment (IQA) methods are utilized to evaluate the performance of the SAR-to-optical image translation quantitatively. The main evaluation metric is the SSIM (Marmanis et al., 2017), which calculates the structural similarity between the generated image and the real image. SSIM will equate to 1 if they are perfectly matched. Another conventional IQA index, peak signal-to-noise ratio (PSNR) operates directly on the intensity over two images (Aljanabi et al., 2019). A higher PSNR corresponds

to a higher quality of the generated image. As well, we compute the root mean squared error (RMSE) between two images (Kulkarni and Rege, 2020). Additionally, we apply the signal-to-reconstruction error ratio (SRE) to measure the error between images of varying brightness (Lanaras et al., 2018).

In the spectral indices computation module, we analyze the probability density functions of the generated spectral indices against real spectral indices and SAR indices. In the wildfire impact assessment, we evaluate the results of threshold-based methods on spectral indices for burned area detection using the Area Under the ROC Curve (AUC) that ranges in value from 0 to 1. A high AUC value indicates a strong separability between burned and unburned classes (Mohajane et al., 2021). In addition, multi-class classification metrics such as precision, recall, overall accuracy (OA), and Kappa coefficient are used for burn severity assessment.

4. Experimental results

4.1. Implementation details

Training the GAN requires a suitable hyperparameter setting to improve stability and reduce the chances of mode collapse (Hughes et al., 2018). To facilitate high-quality reproducibility and examine comparability, we follow the standard settings in the regular Pix2Pix (Zhu et al., 2017) except for hyperparameters in the generator network and loss functions. GAN models are trained from scratch using the PaddlePaddle framework.[2] The batch size is set to 1 by default. Adam optimizer (Kingma and Ba, 2015) is applied to optimize the generator and discriminator network with $\beta_1 = 0.5$. A linear decay learning rate policy is applied with an initial learning rate of 0.0004. The real-time data augmentation on the paired input images includes resizing to 286×286 with bicubic interpolation, random cropping into 256×256, horizontal flipping with 50% probability, random transposing in height and width dimensions with 50% probability.

4.2. SAR-to-optical translation using Pix2Pix

The improved Pix2Pix model differs from previous work in LSGAN and SSIM loss functions and ResNet-based generator architecture. In this section, we investigate the influence of different types of generators and loss functions on the quality of SAR-to-optical image translation.

First, we compare the ground truth against the generated optical images from different Pix2Pix models. The visual comparison results are shown in Fig. 4. The subgraphs of Fig. 4(a) and (f) represent Sentinel-1 SAR image and real Sentinel-2 optical image, respectively. Other images from Fig. 4(b) to (e) are generated optical images by different Pix2Pix models. The generated images by RN6_SSIM_LS have better visual diversity of edges, textures, shapes and colors than other models in different landscapes such as forests, burn scars, lakes, rivers, and bare lands. Despite the spatio-temporal variations in backscattering coefficients, Pix2Pix models can translate SAR images to optical images with rich contextual and semantic information. Specifically, RN6_SSIM_LS model generates fine structured and unstructured textures related to burn scars under the constraint of SSIM loss and LSGAN loss.

For the test patch in a heterogeneous landscape, RN6_SSIM_LS explicitly reconstructs realistic texture in the pre-fire image of CA_SK_2017_1033. It also correctly distinguishes between unburned areas and burned areas with the diverse textures and complex patterns in the post-fire scene of CA_SK_2017_1033. For the test patches CA_BC_2018_182 and CA_MB_2017_904, we can observe that UNet_L1_VA model in Fig. 4(b) may underestimate burned areas, and RN6_L1_VA and RN6_L1_LS models tend to generate compact-shaped burn scars without rich contextual information in Fig. 4(c) and (d). Regarding the pre-fire scenes, RN6_SSIM_LS successfully reconstructs the narrow rivers and streams in CA_BC_2018_182 and small lakes in CA_MB_2017_904.

The quantitative evaluation and analysis were performed for each Pix2Pix model over 1111 test patches, as shown in Table 2. RN6_SSIM_LS achieves the best IQA results with the average PSNR, SSIM, and SRE highest and the average RMSE lowest among all models. The residual generator can lead to a higher IQA than the UNet-based generator. A clear improvement in IQA benefits from the usage of SSIM loss and LSGAN loss.

4.3. Visual interpretation of generated burn severity indices

dNBR, RdNBR, and RBR_{opt} indices provide quantitative estimation of the burned area change between the pre- and post-fire situation (*i.e.,* biomass consumption and related changes) (Lutes et al., 2006; Parks et al., 2014). After obtaining the generated optical images from pre- and/or post-fire SAR images based on the ResNet-based Pix2Pix, we can compute the spectral indices by Eq. (7), (8), and (9). To comprehensively compare the performance between these indices, our experiments are conducted based on four different image combinations:

1. the real pre- and post-fire Sentinel-2 images, namely S2-S2 (RR).
2. the real pre-fire Sentinel-2 image, and fake post-fire Sentinel-2 image generated from Sentinel-1 SAR image, namely S2-S1 (RF).
3. the fake pre-fire Sentinel-2 image generated from pre-fire Sentinel-1 SAR image, and real post-fire Sentinel-2 image, namely S1-S2 (FR).
4. the fake pre-fire Sentinel-2 image generated from pre-fire Sentinel-1 SAR image, and fake post-fire Sentinel-2 image generated from post-fire Sentinel-1 SAR image, namely S1-S1 (FF).

A graphical representation of the metrics calculation module is given in Fig. 5. First, we reconstruct the fake images as shown in Fig. 5(b) from the input SAR images in Fig. 5(a). Then we create the corresponding pre- and post-fire NBR images based on Eq. (6), as shown in the first two columns in Fig. 5(c) after four combinations of bi-temporal scenes. In general, high positive NBR values correspond to vegetated areas, while bare soil and burned areas have low and usually negative NBR values: the lower the NBR values, the higher the burn severity. There is a strong agreement between real NBR (RR) and generated NBR, which provides a basis for us to calculate the spectral indices.

Based on Eq. (7), (8) and (9), the spectral index maps are displayed in the last three columns in Fig. 5(c). Spectral index values increase proportionally to burn severity, and values below zero adequately represent unburned or recovered areas. White areas with higher values probably indicate burn scars. Spectral index maps show very fine-structured burned areas, comparable to the reference combination (RR) in the first row in Fig. 5(c). Due to the different transformation functions, the value range of RBR_{opt} is narrowed, while RdNBR is expanded. Using the same color bar from −1500 to 1500, higher values in RdNBR but relatively lower values in RBR_{opt} represent the burned area. As expected, the mean value of dNBR should be between RdNBR and RBR_{opt} (*cf.* Table 3).

Based on the equivalent data obtained from test wildfire scenes randomly (n = 111,100), the mean value of post-fire NBR is much lower than that of pre-fire NBR in Table 3. Fake pre-fire NBR images in the FR and FF have lower mean values than that of real ones (456.3<473.9), while fake post-fire NBR in the RF and FF have larger mean values than the real ones (120.1>83.2). This phenomenon leads to the lower values of generated indices from RF, FR, and FF compared to the RR in Table 3. Overall, the mean values of dNBR and RBR_{opt} from FR are closer to RR but the mean RdNBR from RF appears to be closer to RR.

[2] https://github.com/PaddlePaddle/Paddle

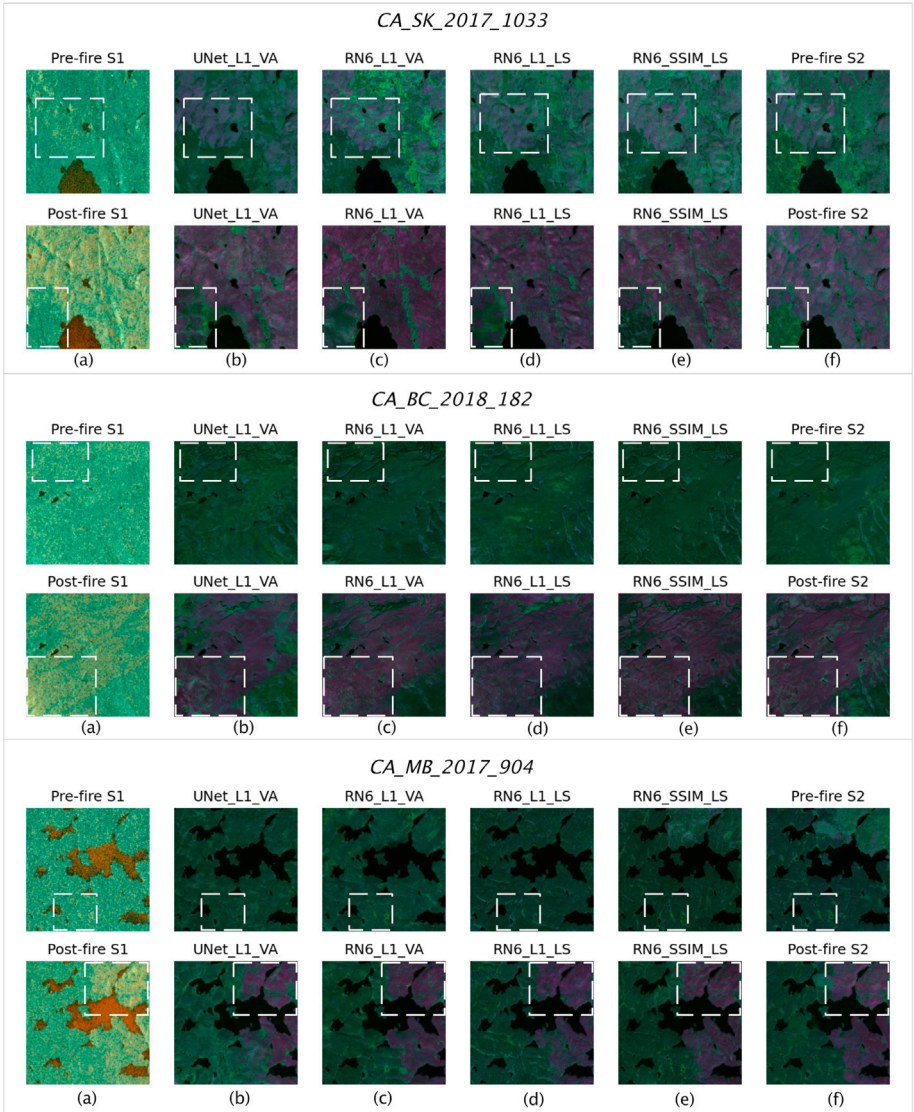

Fig. 4. Visual comparison of SAR-to-optical translation results from different Pix2Pix models applied to the test patches from three provinces of Canada. From left to right: (a) input pre- or post-fire Sentinel-1 image, (b) UNet-based generator with L1 loss and vanilla loss, (c) ResNet6-based generator with L1 loss and vanilla loss, (d) ResNet6-based generator with L1 loss and LSGAN loss, (e) ResNet6-based generator with SSIM loss and LSGAN loss, and (f) real pre- or post-fire Sentinel-2 image.

Table 2
IQA results of different methods.

S1-to-S2	Generator	G-Loss	D-Loss	PSNR↑	SSIM↑	RMSE↓	SRE↑
UNet_L1_VA (baseline)	UNet	L1	vanilla	21.6	0.541	21.6	46.7
RN6_L1_VA	ResNet-6	L1	vanilla	22.0	0.576	20.6	47.0
RN6_L1_LS	ResNet-6	L1	LSGAN	22.1	0.571	20.4	47.0
RN6_SSIM_LS (ours)	ResNet-6	SSIM	LSGAN	**22.2**	**0.593**	**20.3**	**47.1**

Table 3
Mean values and proportion of positive values of the spectral indices in terms of dNBR, RdNBR, and RBR_{opt}.

Group	preNBR	preNBR	postNBR	postNBR	dNBR	RdNBR	RBR_{opt}	
	mean	%positive	mean	%positive	mean	mean	mean	%positive
S2-S2 (RR)	473.9	98.78	83.2	55.33	390.7	553.9	256.5	90.48
S2-S1 (RF)	473.9	98.78	120.1	60.58	353.7	499.9	231.3	88.80
S1-S2 (FR)	456.3	99.07	83.2	55.33	373.1	485.4	248.0	87.80
S1-S1 (FF)	456.3	99.07	120.1	60.58	336.1	446.1	225.1	89.86

4.4. Probability density functions and spatial correlation

Fig. 6 exhibits probability density functions plots of the generated and real spectral indices, as well as their relationships with RBR_{sar} index. We can observe both high similarity and slight differences between generated spectral indices and real ones in probability density functions. The real indices more focus on the zero value while the three generated spectral indices in RF, FR, and FF show slight negative biases

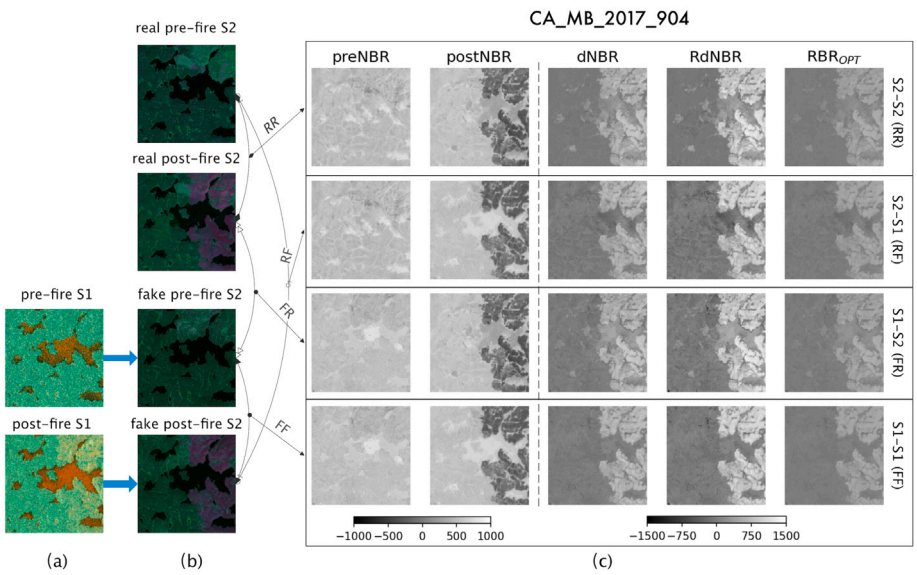

Fig. 5. Visual comparison of generated spectral indices on a test patch. (a) Input Sentinel-1 SAR image. (b) From top to bottom: real pre-fire Sentinel-2, real post-fire Sentinel-2, fake pre-fire Sentinel-2, and fake post-fire Sentinel-2 images. (c) Spectral index maps under different combinations.

as mean values tend to be lower than those in RR combination (*cf.* Table 3). The generated dNBR and RBR$_{opt}$ indices are highly correlated with the real indices. Specifically, dNBR (FR) has a higher correlation coefficient (r = 0.84) than RdNBR (r= 0.27) and RBR$_{opt}$ (r = 0.81). In Fig. 6(b), RdNBR is more sensitive than dNBR and RBR$_{opt}$ owing to the square root transformation of the denominator in Eq. (8) that causes extremely small values of pre-fire NBR (Miller and Thode, 2007). Conversely, the probability density functions comparison of RBR$_{opt}$ and dNBR in Fig. 6(a) and (c) have similarities except for the converged shape of RBR$_{opt}$ owing to its adjustment to the pre-fire NBR by adding 1.001 to the denominator (*cf.* Eq. (9)) (Parks et al., 2014).

We additionally compare SAR-based RBR$_{NDBI}$ with optical spectral indices. RBR$_{NDBI}$ (*cf.* Eq. (10)) has been computed as the average post-fire backscatter over the post-fire period divided by the average pre-fire backscatter over the pre-fire period (Section 2.2), to get rid of the speckle effect and produce smoother textures (Tanase et al., 2015). There is no strong agreement between RBR$_{NDBI}$ and real spectral indices in Fig. 6(d) due to the different imaging mechanisms.

5. Discussion

5.1. Binary classification on generated spectral indices

The spectral indices can be calculated from pre- and post-fire NBR, where negative and lower values represent regrowth and unburned vegetation. Higher values correspond to burn scars. Spectral indices are often calculated using single-source imagery. In this study multi-source observations (*e.g.,* SAR and optical) are adapted to the same domain using the ResNet-based Pix2Pix model to enable retrospective burned area mapping. The major challenge of burned area classification is to determine prescribed thresholds with physical and empirical insights on spectral indices units (Hall et al., 2008). A suitable threshold t ($100 < t < 1500$) over spectral indices can effectively separate the burned pixels from the unburned pixels. Considering different reliabilities of the stand-replacing spectral indices thresholds (dNBR, RdNBR, and RBR$_{opt}$), we evaluated the performance of generated indices in classifying burned areas on test patches using the AUC measure.

AUC values of different combinations are reported in the form of boxplots in Fig. 7; boxes cover the interquartile range (25th percentile to 75th); black whiskers indicate values from the 5th to the 95th percentile; gray lines in boxes indicate the median. Optical combinations (RR, RF, FR, and FF) achieve higher classification performance than SAR-based indices (RBR$_{NDBI}$, RBR$_{VH}$, and RBR$_{VV}$). All generated indices have a 3rd quartile of AUC above 0.8, indicating good classification accuracies for burned areas. In particular, FR shows lower AUC variations between RF, FR, and FF combinations and it also has medium AUC values closer to the RR group. In addition, SAR-based indices perform poorly in detecting burned areas with larger AUC variations and lower medium AUC values. RBR$_{NDBI}$ significantly outperforms RBR$_{VH}$ and RBR$_{VV}$ as NDBI is more sensitive to forest disturbances. Despite showing lower accuracies than the RR group, the generated spectral indices are well-established and much more effective in detecting burned areas than straightforward SAR-based indices.

Table 4
Thresholds for three burn severity metrics.

Metric	Unchanged/low	Low/moderate	Moderate/high
dNBR	160	260	481
RdNBR	230	475	835
RBR$_{opt}$	105	200	355

5.2. Multi-source data fusion for burn severity mapping

Generated spectral indices can highlight burned area changes by continuous values (*cf.* Fig. 5). They can be further classified into discrete severity classes (Parks et al., 2014) to obtain burn severity maps. In this study, the values of each generated spectral index are differentiated into four severity categories (*i.e.*, unchanged, low, moderate, and high) using the analyst-derived thresholds in M.A. and Fernández-Manso (2017) as shown in Table 4. These thresholds have been successfully applied for burn severity mapping with Sentinel-2 data (Quintano et al., 2018).

Table 5 presents the burn severity classification accuracies in RF, FR, and FF combinations. Ground reference data are from the RR group based on thresholds in Table 4. dNBR has a higher OA and Kappa coefficient than RdNBR and RBR$_{opt}$ under RF and FF combinations that occupy a fake post-fire optical image. RdNBR does appear to outperform dNBR index at FR combination dependent on the real post-fire image. RBR$_{opt}$ index is a robust metric for classifying burn severity with an accuracy between dNBR and RdNBR. As discussed by Soverel et al. (2010), RdNBR did not outperform dNBR in burn severity class proportions across western Canada. Our results indicate

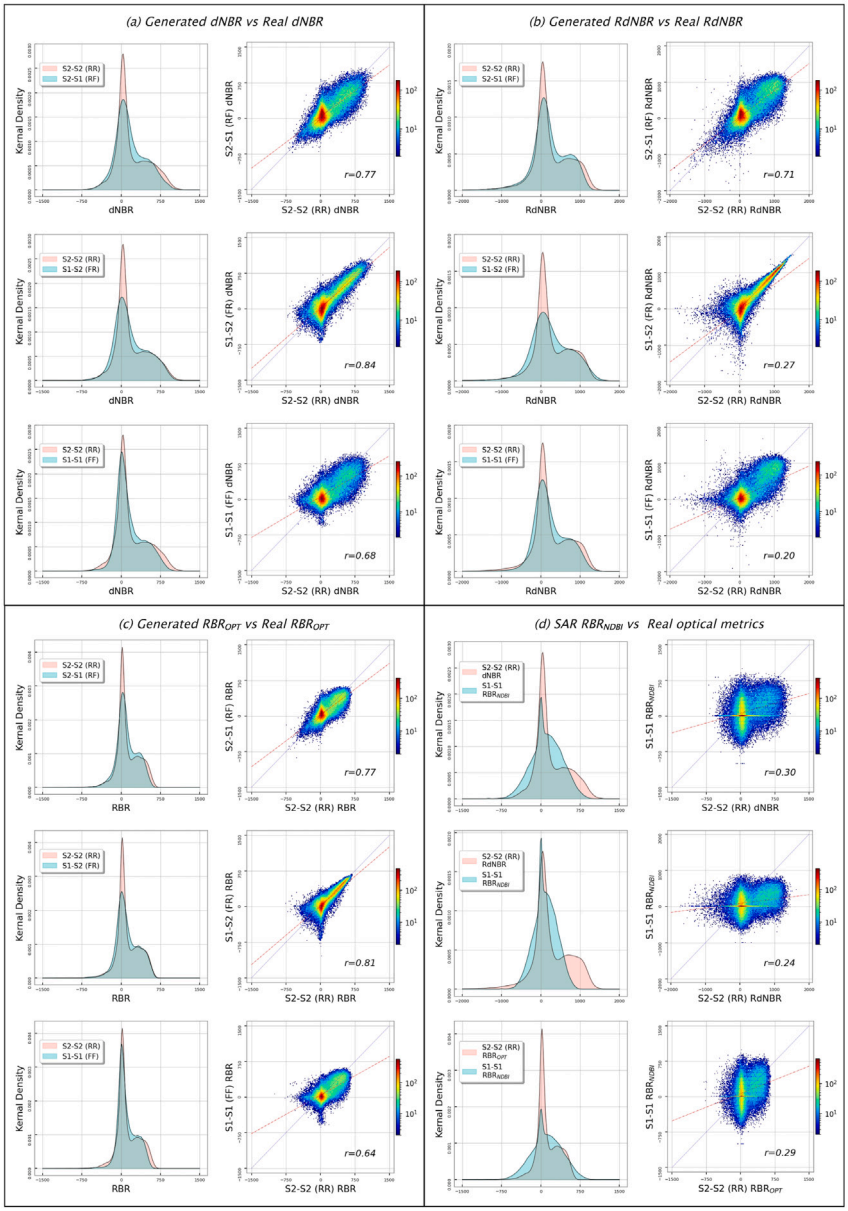

Fig. 6. Gaussian kernel densities and scatterplots showing the relationship between real images (RR) and generated images (RF, FR, or FF) in terms of dNBR (a), RdNBR (b) or RBR_{opt} (c) indices. The equivalent data are obtained from 1111 wildfire test patches (n = 111,100). Additionally, RBR_{NDBI} is compared with the real dNBR, RdNBR, and RBR_{opt} in (d). r is Pearson's correlation coefficient and the color bar signifies the density of points.

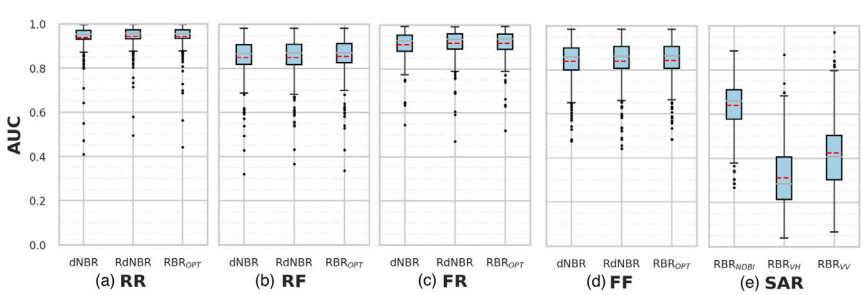

Fig. 7. Boxplots of the AUC values at different combinations for burned area classification on test patches in terms of spectral indices.

Table 5
Burn severity classification accuracies based on the generated indices under different combinations.

Metric		Precision				Recall				F1-score					
		U	L	M	H	U	L	M	H	U	L	M	H	OA	Kappa
dNBR	RF	0.77	0.15	0.35	0.75	0.86	0.19	0.37	0.62	0.79	0.15	0.35	0.62	**63%**	**0.44**
	FR	0.86	0.28	0.56	0.83	0.92	0.29	0.55	0.79	0.88	0.28	0.54	0.78	76%	0.63
	FF	0.72	0.13	0.32	0.71	0.84	0.18	0.34	0.54	0.75	0.14	0.32	0.56	**59%**	**0.39**
RdNBR	RF	0.76	0.22	0.40	0.62	0.85	0.27	0.40	0.47	0.78	0.22	0.38	0.46	56%	0.37
	FR	0.88	0.55	0.81	0.90	0.94	0.55	0.78	0.88	0.90	0.54	0.78	0.88	**84%**	**0.77**
	FF	0.73	0.21	0.40	0.62	0.85	0.27	0.39	0.45	0.76	0.22	0.37	0.45	55%	0.35
RBR_{opt}	RF	0.76	0.19	0.37	0.68	0.86	0.24	0.37	0.55	0.79	0.20	0.35	0.54	58%	0.40
	FR	0.87	0.43	0.68	0.85	0.93	0.44	0.66	0.82	0.89	0.43	0.66	0.81	80%	0.69
	FF	0.72	0.18	0.36	0.67	0.85	0.24	0.36	0.50	0.76	0.19	0.34	0.51	56%	0.37

OA: overall accuracy; U: unburned; L: low burn severity; M: moderate burn severity; H: high burn severity. **Bold**: burn severity classification results with the highest OA and Kappa for each combination.

that this hypothesis may be acceptable for Canadian fire events. For GAN-based burn severity mapping, dNBR index is recommended to adopt when post-fire optical images are not available while RdNBR is suitable in conditions without available pre-fire optical images.

Fig. 8 shows the visual burn severity maps over a test patch and the Gaussian kernel density functions between real spectral indices and generated indices over this test patch. Different thresholds for spectral indices can lead to various classification maps even in the RR combination. More pixels in the dNBR-based severity map are classified into the high severity class, but RdNBR is more sensitive to the low severity. By visual interpretation, the FR shows higher agreement with the RR. In Fig. 8(d), FR also shows a higher correlation in probability density functions than RF and FF combinations. There is a distribution discrepancy between RF or FF and RR. The generated images for burn severity mapping inherently present a lower performance than real optical images due to the heterogeneous landscape in real scenarios and the lack of diversity of the generated images. However, as an alternative to assessing damage severity with SAR data, generated burn severity indices can facilitate the description and study of post-fire damage patterns.

5.3. GAN-based multi-source fusion

The proposed GAN-based framework for assessing wildfire impacts with multi-source satellite images is a new contribution to wildfire research and science. In particular, our methodology can work under all-weather conditions to quickly generate burned area perimeters and burn severity products as an emergency response after natural disasters. In addition, the SAR-to-optical framework allows those who are not SAR experts but are familiar with optical image processing to assess wildfire impacts based on the generated spectral indices from available SAR data. Experimental results indicate that bi-temporal spectral indices based on generated optical images show similar trends to real optical images over a range of fire regimes in Canada.

In practical applications, we often encounter challenges in making use of multi-source data, especially SAR data. Conventional SAR-optical fusion methods based on deep learning models mostly focus on stacking radar-derived images as well as optical composites (*e.g.,* spectral bands and spectral indices) and then feeding them into CNN networks for burned area mapping (Belenguer-Plomer et al., 2021; Rashkovetsky et al., 2021). These wall-to-wall data-driven approaches have been widely used in remote sensing (Sharma et al., 2018; Tavares et al., 2019; Scarpa et al., 2018), but mostly SAR data did not lead to satisfying results, or these models did not take advantage of SAR data. The main reason may be that heterogeneous optical and radar images would confuse CNN networks to exhibit good generalization. Even worse, these neural networks may underachieve or collapse under the contradiction between the different imaging mechanisms of SAR and optical sensors. Stacked data with radar interferometric coherence also increases the computational complexity (Zhang et al., 2021).

Some studies have shown that SAR-based change detection for burned areas is quite promising but still needs further improvement compared to optical image (Dice 0.57 *vs.* 0.7) (Belenguer-Plomer et al., 2021). By translating SAR images into optical images using the proposed SAR-to-optical translation framework, generated images exhibit spectral information representing the burned area changes without the effect of speckle noise in the SAR images (Li et al., 2021). Our experimental results show that spectral indices generated from pre- or/and post-fire SAR images can outperform the pure SAR-based index in burned area mapping (*cf.* Fig. 6). The generated optical indices can compensate for the drawback of few available optical images in cloudy regions, which allows for the application of simpler and more robust empirical solutions (such as threshold-based approaches) to detect burned areas or estimate damage levels without needing costly field data.

Fusion of multi-source data will substantially increase the temporal observation frequency (Chastain et al., 2019). Landsat-8, Sentinel-2A, and Sentinel-2B together can provide a global median average revisit of 2.9 days (Li and Roy, 2017). Adding longer wavelengths may allow for discriminating burn scars in forested areas, difficult to be detected from optical and shortwave SAR wavelengths (Belenguer-Plomer et al., 2021). Hence, the proposed GAN-based framework may greatly benefit from the use of different SAR wavelengths (Sentinel-1 in the C band, PALSAR-2 in the L band, and future Biomass in the P band) and multispectral data (Landsat-8, Landsat-9, and Sentinel-2) with the fine resolution. This suggests promising future work toward the multi-source translation between the multi-frequency SAR sensors and multi-resolution optical sensors for burned area detection and burn severity assessment.

6. Conclusion

SAR is an important alternative data source for monitoring fire-prone regions under thick clouds or smoke. In contrast to widely used optical spectral indices (*e.g.*, dNBR), the interpretation of SAR images on burned areas remains challenging due to the spatio-temporal variability (*e.g.,* different soil moisture levels, multiple orbit acquisitions, burned vegetation characteristics). On the other hand, traditional spectral-rule algorithms that are successfully applied in optical images cannot be easily adapted to SAR data due to disparate imaging mechanisms. To bridge the gap between SAR and optical data, this study proposed an effective framework that combines a SAR-to-optical translation network and a spectral indices calculation module to handle multi-source data for burned area detection and burn severity mapping. The proposed framework can leverage both SAR and optical data in case optical observations are not available in areas with persistent cloud cover or thick smoke.

The analysis was carried out on 8669 pairs of bitemporal SAR and optical patches over 304 large wildfire events (223,578 km^2 study areas) in Canada with a wide range of land covers. Based on the ResNet-based Pix2Pix with the SSIM-based reconstruction loss and LSGAN

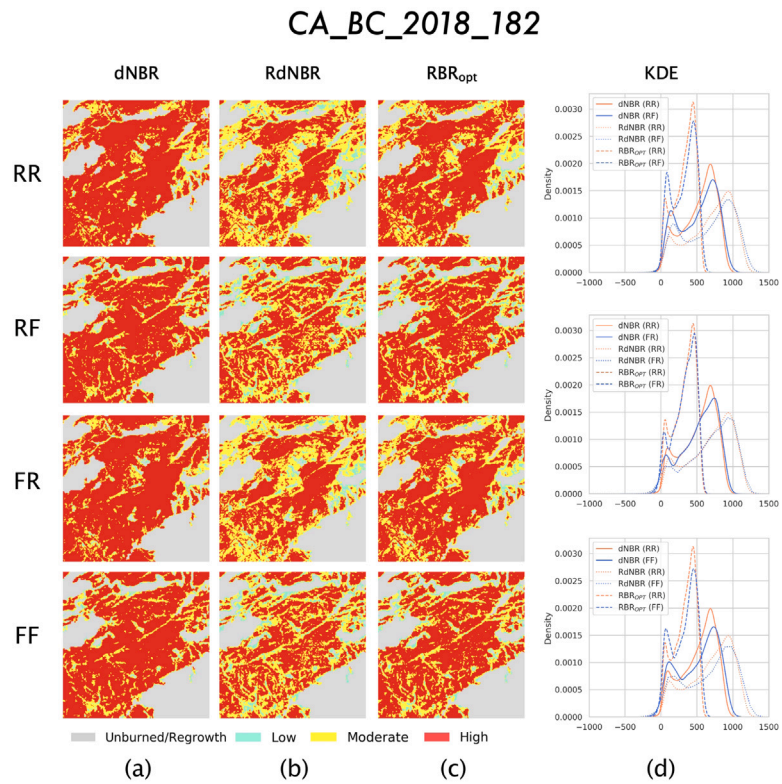

Fig. 8. Burn severity classification maps based on dNBR (a), RdNBR (b) and RBR_{opt} (c). (d) Gaussian kernel density estimate maps showing the distribution of spectral index values between RR and RF, FR, or FF.

adversarial loss, we stabilized the training of SAR-to-optical translation model and improved the quality of the generated images with increased SSIM from 0.541 to 0.593 and better IQA. By performing a correlation analysis, we found that the generated indices have a much higher correlation with the real indices than the original SAR-based indices. Furthermore, burned area detection and burn severity classification were performed and discussed under different multi-source combinations to validate the effectiveness and universality of the generated indices.

This study provides insights into the GAN-based SAR and optical image translation for wildfire impact assessment using Sentinel-1 and Sentinel-2 data, which allows us to (i) represent the change in radar backscatter using spectral information, (ii) apply traditional spectral indices to SAR data, and (iii) improve burned area classification accuracy of SAR-based index. Despite these strengths, GAN-based translation algorithms for burned area detection are limited by several sources of errors, including unbalanced training data between burned and unburned areas, heterogeneous landscapes, the chances of model collapse, and the presence of land changes not caused by wildfire. Future work would focus on improving the robustness of image translation from both model and data aspects.

CRediT authorship contribution statement

Xikun Hu: Conceptualization, Investigation, Data curation, Methodology, Visualization, Validation, Writing – original draft, Writing – review & editing. **Puzhao Zhang:** Resources, Data curation, Methodology, Writing – review & editing. **Yifang Ban:** Conceptualization, Investigation, Writing – review & editing, Supervision, Funding acquisition. **Maryam Rahnemoonfar:** Investigation, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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